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# Artificial Intelligence Measurement of Disclosure (AIMD)

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# Artificial Intelligence Measurement of Disclosure (AIMD)

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ABSTRACT Empirical research on voluntary disclosure lacks an appropriate measurement technique for quantifying the intensity of a firm's disclosure. In this paper, I introduce artificial intelligence measurement of disclosure (AIMD), a computerised technique for measuring disclosure using artificial intelligence, which derives disclosure proxies from English-language annual reports for 10 different information dimensions without human involvement. Criterion validity tests indicate that, controlling for a robust set of covariates and multiple statistical techniques, AIMD is negatively associated with information asymmetry as proxied by spreads and PIN. Furthermore, AIMD has construct validity when compared to the AIMR disclosure rating, Standard & Poor's Transparency and Disclosure Rating, several proprietary manual disclosure scorings and companies' own assessment of their level of disclosure as indicated by a survey. I also demonstrate the applicability of AIMD as a cost-effective technique for measuring disclosure using a sample of 127,895 firm-year observations of companies regulated by the SEC.

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## 1. Introduction

Existing research into corporate disclosure takes a number of different approaches ranging from earnings-based techniques to multidimensional techniques looking at sales, R&D, corporate strategy and other areas (see Healy and Palepu, 2001 for a comprehensive review). No generally accepted and broadly applicable measurement technique exists which would allow researchers to evaluate the extent of a company's disclosure. Moreover, empirical research often suffers from small sample sizes due to the considerable effort involved in carrying out scorings manually. In addition, the availability of secondary data

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is very limited, with the result that samples are often more than a decade old and heavily biased towards large companies.

These limitations have led to calls for computerised techniques for measuring disclosure to be developed (Core, 2001, p. 452). This paper attempts to provide such a measurement technique: AIMD, an artificial intelligence computerised measure of disclosure. AIMD measures the extent of corporate disclosure in English-language documents such as companies' annual reports. It does so along 10 information dimensions: (1) information about sales markets and customers; (2) information about employees; (3) information about the corporate environment; (4) financial information; (5) information about corporate governance; (6) information about R&D; (7) information about social and environmental responsibility; (8) information about capital markets; (9) information about corporate strategy; and (10) information about the value chain.

AIMD differs from existing techniques for measuring disclosure in a number of ways. It looks at many different aspects of disclosure, including non-financial and voluntary disclosure. It does not analyse individual words, but more complex semantic units (phrases) and to some extent the grammatical structure of sentences. It does not require human interaction or human judgement to measure the extent of disclosure in the annual reports. Therefore, AIMD is cost effective and can be used with large samples.

AIMD uses an information-retrieval vector space model based on the techniques of artificial intelligence (AI). It consists of the two usual phases of AI: training and application. In the training phase, a coding scheme is created on the basis of a sample of representative annual reports, attempting to classify and quantify the diversity of corporate disclosures. The resulting coding scheme comprises 13,314 standardised items, covering about $10^{24}$ different semantic units relevant for corporate disclosure. In the application phase AIMD then applies the coding scheme to a larger number of annual reports without human involvement.

In this paper, I also examine how reliable – and valid – the measurements produced by AIMD are. I test criterion validity by looking at whether the measurements produced by AIMD are negatively related to various information asymmetry proxies: a significant negative association is found between the bid-ask spread and AIMD for a sample of 13,710 firm-year observations. Similarly, the probability of information-based trading (PIN) is found to be significantly negatively associated to AIMD for a sample of 11,606 firm-year observations. Construct validity is demonstrated against the Association for Investment and Management Research (AIMR) disclosure rating, Standard & Poor's Transparency and Disclosure Rating, the disclosure ratings of two business magazines and companies' own assessments of their level of disclosure as indicated by a survey using samples between 85 and 528 firm-year observations.

As well as annual reports, AIMD can use quarterly reports, the minutes of meetings with analysts, press releases, press clippings, records of conference

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calls, and company websites. Looking at a sample of 127,895 firm-year observations from between 1996 and 2008 for companies regulated by the Securities and Exchange Commission (SEC) (see Appendix D), I show that AIMD is a cost-effective technique for quantifying the level of disclosure and a suitable tool for large-scale empirical research.

This paper contributes to research in the field of corporate disclosure in a number of ways. Previously, the use of content analysis techniques in research into corporate disclosure was limited to techniques based on manual scoring (e.g. Cooke, 1989; Meek *et al.*, 1995; Botosan, 1997; Guthrie *et al.*, 2004; Beattie *et al.*, 2004; Vandemaele *et al.*, 2005; Santema *et al.*, 2005; Cheng and Courtenay, 2006) and individual word counts, generally done by computer (e.g. Ingram and Frazier, 1983; Frazier *et al.*, 1984; McConnell *et al.*, 1986; Taylor and Ingram, 1984; Tennyson *et al.*, 1990; Abrahamson and Amir, 1996; Smith and Taffler, 2000; Ahmad *et al.*, 2003; Hussainey *et al.*, 2003; Mauldin and Richtermeyer, 2004; Rutherford, 2005; Beattie and Thomson, 2007; Kothari *et al.*, 2009; Loughran and McDonald, 2011; Demers and Vega, 2010). Both of these approaches are heavily criticised in the literature (e.g. Marston and Shrives, 1991; Beattie and Thomson, 2007). AIMD responds to these criticisms by using artificial intelligence rather than manual scoring, and by looking not at single words but at more complex semantic units and to a certain extent grammar.

AIMD also makes it possible to re-examine the findings of earlier research by testing them against larger sample sizes, a greater number of covariates and more tests for robustness. Furthermore, it can be used to test a number of hypotheses at the same time, such as the association between disclosure and particular governance structures and their interaction with capital market attributes (analyst following, regulatory environment, litigation environment, etc.). Researchers have examined these relationships individually in the past (e.g. Welker, 1995; Leuz and Verrecchia, 2000; Chen *et al.*, 2007) but AIMD opens the door to more sophisticated tests using larger sample sizes and avoiding the bias arising from omitted variables.

Finally, AIMD also allows disclosure time series to be derived (see Appendix D). This can shed light on how disclosure has developed over specific periods of time (see, for example, Hutajulu and Fogarty, 2006). It may also help to enhance international comparative disclosure research (e.g. Roberts, 1990, 1991; Meek *et al.*, 1995; Adams *et al.*, 1995, 1998; Adams and Kuasirikun, 2000).

This paper is structured as follows: Section two gives a brief outline of previous approaches to measuring corporate disclosure; in Section three I introduce the AIMD measurement technique; in Section four I evaluate the reliability and validity of AIMD; and, finally, in Section five I present my conclusions.

## 2. Approaches to Measuring Disclosure

Three basic types of approaches to measuring disclosure exist. *Sender approaches* quantify disclosure based on self-evaluation by companies; rare

examples using questionnaires include Günther *et al.* (2003) and Deegan and Blomquist (2006). *Receiver approaches*, by contrast, measure disclosure based on the professional judgement of financial analysts. A well-known example is the disclosure ranking published by the Association for Investment and Management Research (AIMR; see Lang and Lundholm, 1993; Welker, 1995; Lundholm and Myers, 2002; Gelb and Greenstein, 2004; Pronk, 2006). The most common type of approach, however, is *observer approaches*. These approaches analyse texts or documents produced by companies, such as annual or quarterly reports.

The first two types of approach have significant limitations. Sender approaches, for example, have limited applicability due to economic reasons. Healy *et al.* (1999) and Healy and Palepu (2001) likewise raise serious concerns about receiver approaches. In the following discussion I therefore focus on observer approaches that analyse the content of texts.

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### *2.1. Content Analysis Approaches*

The basic idea behind *content analysis* – a technique widely used in social sciences (Krippendorff, 2004) – is to reduce the amount of data in a text by classifying words, phrases, sentences, paragraphs and so on into a smaller number of 'content categories' (Weber, 1990, p. 15). Approaches based on this technique measure the frequency of occurrence of semantic units such as topics, issues, phrases or words in a text (Weber, 1990, p. 72). Rules are established determining what content category each semantic unit belongs to; these 'assignment rules' are then used to analyse the content of a specific text.

Morris (1994, p. 904) distinguishes three types of content analysis approaches: human-scored schemes, individual word-count approaches and content analysis approaches using AI.

*Human-scored schemes* classify whole texts, paragraphs or sentences into different categories by having one or more human beings 'score' them. Checklists are frequently used to increase the reliability of the scoring. Examples include Cooke (1989), Meek *et al.* (1995), Botosan (1997), Guthrie *et al.* (2004), Beattie *et al.* (2004), Vandemaele *et al.* (2005), Santema *et al.* (2005) and Cheng and Courtenay (2006). Computers can also be used to assist the scorers (e.g. Beattie *et al.*, 2004). Due to the considerable effort required in categorising texts, checklist-based approaches often use only small samples. Moreover, results can be difficult to replicate due to limited intertemporal and interpersonal reliability, that is, the subjectivity of the coding. The validity of checklist-based approaches has also been strongly criticised for its subjective selection of criteria and weighting of the elements on the checklists (Marston and Shrives, 1991, pp. 197–199).

Morris's second type of content analysis approach – *individual word-count approaches* – works on the assumption that texts with similar key terms deal with similar topics. Consequently the more frequent these words are, the more important they are for the contents of the text (Luhn, 1959a, 1959b, 1959c;

Carroll and Roeloffs, 1969, p. 228; Berry, 2004, p. 106). For example, the more often the words 'auditor', 'cost' or 'equity' appear in a text, the more likely the text is to be about accountancy. Word frequency was suggested as a proxy for corporate disclosure as early as the 1960s (Copeland and Fredericks, 1968, p. 109). Nevertheless, this approach has never been as popular as human scoring in research into corporate disclosure. Recent examples of research employing this approach include Smith and Taffler (2000), Ahmad *et al.* (2003), Hussainey *et al.* (2003), Mauldin and Richtermeyer (2004), Rutherford (2005), Beattie and Thomson (2007) and Kothari *et al.* (2009).

Morris's third type of content analysis approach – approaches using *artificial intelligence (AI)* – is related to word-count approaches, but additionally considers syntax and lexicon (Rosenberg *et al.*, 1990, p. 299). These approaches do not look at the frequency of words but at larger semantic units (phrases, sentences) (Weber, 1990, p. 72). They generally require some kind of knowledge base in order to recognise what the units refer to. Nevertheless, the ultimate goal of fully 'understanding' texts cannot be achieved with current technology (Akerkar, 2005, p. 9; Haddad and Yaseen, 2005, p. 148; Bramer, 2009, p. 15). As far as I am aware, AIMD is the first attempt to use a content analysis approach based on AI to examine corporate disclosure.

Research has shown that the quality of content measurement varies between the different approaches. Deffner (1986) argues that 'reliability' and 'validity' are inversely related, with word-count systems showing the greatest reliability and human-scored approaches the greatest validity. Hussainey *et al.* (2003) argue that the validity of word-count approaches is limited because of the varying relevance of coding units (some words or sentences might be more important than others), because of the focus on textual documents, and because of company-specific factors such as its industry, segments and so on.

Rosenberg *et al.* (1990) conclude that computerised methods are superior to human-scored approaches and various computerised approaches do not differ significantly in their classification performance. In a related vein Hussainey *et al.* (2003) analyse annual report disclosure and report that human scorings and word-count approaches find congruent results for 85% of the sentences analysed. The correlation coefficient between both disclosure measures is 0.96 and highly significant.

By contrast, Beattie and Thomson (2007), in their detailed analysis of a single annual report, find that only 17.3% of the information about intellectual capital identified by human scoring could be confirmed by a word-count approach. Beattie and Thomson (2007, p. 157) admit that 'in some cases one electronic hit provided more than one piece of … information', but about 40% of word-count hits contained no information at all about intellectual capital. The validity of the word-count approach might improve if a thesaurus were used or if the items on the wordlists were derived from actual disclosure texts rather than a content analysis of literature on intellectual capital. Nevertheless, serious concerns remain about computerised measures of disclosure.

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Accordingly, the reliability and particularly the validity of content analysis approaches need to be reviewed carefully. As a consequence, I comprehensively analyse the reliability and validity of AIMD in Section 4.

A further problem with existing research into corporate disclosure using content analysis is that it often fails to state explicitly whether it is investigating the *quantity* or the *quality* of the disclosure (Beretta and Bozzolan, 2008). Most existing research uses some kind of quantity measure, counting the occurrence of certain items. This reflects the fundamental assumption of content analysis that content is correlated to the occurrence of certain characteristic terms (Weber, 1990). Techniques for measuring disclosure which claim to measure the quality of disclosure, on the other hand, vary with respect to the information dimensions they (additionally) consider (e.g. time, historical vs. forward-looking information, financial vs. non-financial information, qualitative vs. quantitative information) and the aggregation rules (e.g. concentration measures) they apply (e.g. Beattie *et al.*, 2004; Beretta and Bozzolan, 2004). Of course, in reality these measurement techniques make the same assumption about the relationship between content and the occurrence of certain characteristic terms.[1] As 'quantitative' and 'qualitative' disclosure measures both identify disclosure-relevant terms in a text but only differ in the set of characteristic terms and aggregation rules I do not draw on the arbitrary distinction between quantity and quality measures in the discussion below.

### *2.2. Computer-Assisted Content Analysis*

Several computer-assisted content analysis techniques are described in the literature. Popping (1997) provides a comprehensive review of existing software packages. In this section I review a number of recent applications so as to highlight the contribution of AIMD to empirical research into corporate disclosure.

Schnatterly (2003) uses the contents of annual reports as a proxy for operational governance using NUD*IST. The content measure is a manually assigned score based on the 10 lines of text before and after the occurrence of a word featured on a list of 36 words. Similarly Beattie *et al.* (2002, 2004) use NUD*IST to support manual scoring in certain annual reports. Unlike many other studies they address validity and reliability issues in detail.

Abrahamson and Amir (1996) use software to support manual coding and to carry out a word-count analysis based on a list of 56 words indicating 'bad news'. Unfortunately the authors do not state to what extent the findings of the two analyses differ.

Beattie and Thomson (2007) compare human-scored schemes with word counts for disclosure about intellectual capital in the Next plc 2004 Annual Report. The word-count measure uses a wordlist of 105 items relating to different sub-categories of intellectual capital. Based on their analysis, the authors criticise word-count approaches for their low level of validity.

Kothari *et al.* (2009) take a two-stage approach. They first create six wordlists relating to different content categories in corporate reports, analysts' reports and

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press articles, each containing between 26 and 63 words. Using the wordlists, Kothari *et al.* (2009, p. 1649) then 'cluster disclosure texts with similar words and word meanings together'. Unfortunately they do not provide any detailed documentation relating to this process. Next they measure the extent to which texts contain favourable or unfavourable information using a word-count approach based on the General Inquirer's positive and negative wordlists.

Similarly, Hussainey *et al.* (2003) identify topics relating to profit using an individual word-count analysis based on 12 words (including plurals where appropriate). They also develop an alternative, more comprehensive wordlist of some 500 items reflecting a broader understanding of corporate disclosure and identify forward-looking statements in annual reports based on a list of 59 items. This approach is also employed by Schleicher *et al.* (2007) and Hussainey and Walker (2009).

To summarise, two general approaches to computerised content analysis are found in the literature. In the first, the researchers use software to support human scoring procedures and thereby reduce the manual effort involved (e.g. Abrahamson and Amir, 1996; Beattie *et al.*, 2002, 2004; Schnatterly, 2003). Nevertheless human interaction is still required, which has a negative impact on reliability and inevitably restricts sample sizes. In the second approach, the researchers use computer software to perform a content analysis without human involvement by means of word-count approaches based on limited lists of keywords (e.g. Abrahamson and Amir, 1996; Beattie and Thomson, 2007; Kothari *et al.*, 2009). However, it is impossible to address broad disclosure topics with small keyword lists and the applicability of the findings is limited due to serious concerns about validity.

AIMD differs from the above methodologies in several respects. It does not require human interaction and judgement, and is thus not subject to reliability issues. It covers a broad range of disclosure topics – including non-financial voluntary disclosure – across 10 information dimensions. Furthermore, it does not analyse the frequency of single words, but uses AI techniques to analyse more complex semantic units (phrases) and to some extent also grammar; it thus partly addresses the context of sentences, too.

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## 3. AIMD

In this section I introduce AIMD, a computerised content analysis approach to measure disclosure based on artificial intelligence, which uses an abstract world model. Following this brief presentation I discuss the theoretical framework of the model and document its implementation in detail.

### *3.1. Overview*

AIMD measures how informative a corporate text is across several information dimensions using an information-retrieval vector space model. The technique

consists of two phases. First, in a training phase, a coding scheme is established based on a sample of annual reports (see Section 3.3). Second, in an application phase, AIMD applies the coding scheme to a larger number of new texts (see Section 3.4).

AIMD reflects a broad understanding of corporate disclosure, including voluntary, non-financial disclosure. Its 10 information dimensions are based on actual disclosure by firms and the results of interviews with financial analysts and academic experts. To this end, annual reports from 2002 published by almost all CAC 40 companies in France, all the DAX-30 companies in Germany and around 30 of the FTSE 100 in the UK in English have been manually examined. Disclosure topics are identified on the basis of the chapter and section headings appearing in these annual reports. In addition financial analysts and other researchers have been interviewed to discuss with them the extensive list of topics using the 'think-aloud method' (Van Someren *et al.*, 1994). This resulted in some new topics being added, but many others being combined. The result was 10 information dimensions covering all relevant aspects of traditional and non-traditional disclosure: (1) information about sales markets and customers; (2) information about employees; (3) information about the corporate environment; (4) financial information; (5) information about corporate governance; (6) information about R&D; (7) information about social and environmental responsibility; (8) information about capital markets; (9) information about corporate strategy; and (10) information about the value chain. Table 1 outlines the information content of each dimension.

For most of the information dimensions it is not possible to discriminate between voluntary and mandatory disclosure (Cooke and Zeff, 2007). Mandatory disclosure levels also vary between countries and markets. For this reason AIMD does not distinguish between mandatory and voluntary disclosure. This does not weaken the explanatory power of the results as a fixed, mandatory disclosure component will not affect the estimated coefficients except in the form of a constant. However, it might be interesting to see whether one could determine countries' mandatory disclosure levels empirically based on the minimal disclosure provided by a sample of companies.

I chose annual reports to extract the coding scheme for three reasons. First, annual reports are the most important form of corporate disclosure (Courtis, 1986, p. 1; Guthrie *et al.*, 2004, p. 283). Second, they are readily available on companies' websites and other databases in machine-readable form. And third, the other studies of disclosure used in my validity tests (Section 4) also look at annual reports. It is worth noting, however, that AIMD can also be used to analyse other written forms of corporate disclosure, including quarterly reports, company websites, transcripts of analysts' meetings and articles in the press, without prior modification.

Corporate disclosure is a suitable object for computerised content analysis because the audience is cohesive and the knowledge and experience of the

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**Table 1**. Disclosure dimensions: information content and examples of *N*-grams

| Disclosure dimension | # | Information content | Examples of non-standardised *N*-grams |
|---|---|---|---|
| Financial information | (01) | Financial position, profitability, assets, liabilities, equity, liquidity, capital, cash flows, profit distribution, investments | Cash inflow from investing activities; deferred tax liabilities; goodwill amortisation; international accounting standard; total capital spending |
| Sales market and customers | (02) | Sales market, types of customers, competitive position, products/product groups, business segments, marketing, sales | Brand architecture; customer demand; main competitors; marketing channel; position in growth markets |
| Value chain | (03) | Production, logistics, location, procurement, resource efficiency, supply chain | Non-manufacturing activities; order backlog development; key business process; overall production capacity; infrastructure management |
| Employees | (04) | Occupational and operational safety, recruitment, workers' participation, further education | Headcount reduction activities; apprenticeship contracts; social insurance benefits; performance-related compensation components; employee education |
| Research and development | (05) | Research and development activities, new product development, innovation, technology | Cutting-edge technologies; innovative products; new research centre; phase III clinical testing; research area |
| Corporate strategy | (06) | Corporate strategy, strategic goals, risk, acquisitions | Corporate realignment; long-term corporate policy; sales and cost synergies; strategic alliance partners; strategic core business |
| Corporate governance | (07) | Supervisory board, board of management, other boards, corporate governance code, board compensation, auditing | Auditor independence; chairman of the supervisory board; corporate governance; directors' dealing; presidential committee |
| Capital markets | (08) | Capital market situation, investor relations, stock(s), stock performance, dividends, financial instruments | Average number of shares; convertible bonds; cost of equity; derivative financial instruments; price-earnings ratio |

(*Continued*)

**Table 1.** Continued

| Disclosure dimension | # | Information content | Examples of non-standardised *N*-grams |
|---|---|---|---|
| Corporate environment | (09) | Industry situation and development, business cycle, corporate environment and general corporate framework, overall economy and its development | Currency fluctuations; global economic conditions; demographic developments; regional economic activity; trade balance |
| Social and environmental responsibility | (10) | Natural environment protection, ecology, sustainability, charity, health protection, human rights | Clean air measures; cultural sponsorship; equal opportunities activities; renewable energy; socially responsible |

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authors – plus the wording that they use – remains more or less constant. AIMD only uses reports published in English for a number of reasons: in most countries firms are obliged by stock-exchange regulations to publish English translations of most of their corporate reports, and these translations are thorough and accurate (Campbell *et al.*, 2005, pp. 343–344); the grammar of English is easier to analyse than that of many other languages; and suitable computerised dictionaries for English are readily available.

### *3.2. Theoretical Framework*

In this section I briefly outline the concepts and models which form the theoretical framework of AIMD. A fundamental term used here is the '*N*-gram': an ordered sequence of *N* items. In this case the items are words. A single word is a monogram, 'order backlog' and 'backlog order' are examples of bigrams, and 'order backlog development' is a trigram. *N*-grams can be seen as semantic units.

A good way of understanding AIMD is to view it in the context of the information-retrieval vector space model developed in the 1960s (e.g. Salton *et al.*, 1975). In this model, a document is mapped onto a vector space. The components of the text-vector are term frequencies; the cardinality of a finite set of terms determines the space's dimension.

AIMD differs in a number of respects from the many information-retrieval approaches which take single words as their terms (Dominich, 2001, p. 101). In the case of AIMD, the terms are not single words but word-based *N*-grams, that is, sequences of words. Moreover, the set of terms is restricted to those *N*-grams which are relevant for corporate disclosure, that is, which relate to one of the 10 disclosure dimensions, not the entire vocabulary of English. Furthermore, I assume that the meaning of all *N*-grams related to a particular disclosure

dimension is identical, that is, that these *N*-grams are what we might call 'connotative categorical equivalents' – an assumption which reduces the dimension of the vector space considerably. AIMD maps a vector space of standardised *N*-grams with 13,314 dimensions onto the 10-dimensional vector space of the disclosure dimensions. Taking into account certain linguistic standardisations (see Appendix C), it thus covers about $10^{24}$ different *N*-grams relevant for corporate disclosure in company texts.

### *3.3. Training Phase*

For the content analysis (application phase), a coding scheme is required. This is developed in the training phase. The coding scheme describes under what circumstances what disclosure dimension is addressed in a text.

In the training phase a list of all semantic units (*N*-grams) is prepared which are 'connotative categorical equivalents' for a particular disclosure dimension. The result was a separate list of *N*-grams with varying orthography (e.g. 'labor' vs. 'labour') and grammar (plural forms, inflected forms and different word orders within an *N*-gram) for each disclosure dimension. Table 1 gives examples of *N*-grams for each disclosure dimension.

The *N*-gram lists then need to be standardised into a canonical morphological form by treating variation consistently. These standardised *N*-gram lists form the basis for the coding scheme. The coding scheme itself maps all syntactical permutations and orthographic and inflectional variants of *N*-grams onto the disclosure dimensions, ignoring meaningless 'stopwords'.

To this end the *N*-grams are extracted from 24 annual reports representing a very broad range of corporate disclosure. Appendix A describes how I selected these reports, Appendix B how the lists of *N*-grams are drawn and Appendix C how I transformed them into standardised lists. Figure 1 summarises the entire training phase.

It was then necessary to check that the standardised *N*-gram lists were complete – that is, they covered all the *N*-grams relating to the disclosure dimension in question – or if in fact more than 24 annual reports were required for the training phase. I evaluated completeness by looking at the marginal benefit of using additional reports. Table 2 shows the marginal benefit of using one additional annual report for the standardised *N*-gram lists. For example, if the coding scheme were based on a single annual report, the standardised *N*-gram lists would contain 4024 items (i.e. 30.2% of the items based on 24 reports). The contribution to individual disclosure dimensions varies between 844 items (39.1%) for *financial information* and 223 items (22.7%) for *social and environmental responsibility*. If a second report is used, it contributes a further 1869 items (i.e. 14.04% of the items based on 24 reports). The contribution to individual dimensions varies between 334 items (15.48%) for *financial information* and 99 items (15.76%) for *corporate governance information*. Thus the marginal benefit of using a second annual report is substantially lower.

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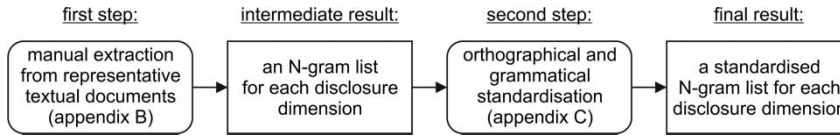



**Figure 1.** Training phase: creating the coding scheme

Overall, the marginal benefit of using additional annual reports falls considerably with each additional report used. For example, the total marginal contribution of the eighteenth annual report is less than 1%; the twenty-fourth report contributes just 0.35% of the total items, and does not contribute to two disclosure dimensions at all. I therefore consider 24 reports to be sufficient for creating a complete coding scheme.

### *3.4. Application Phase*

In the application phase, AIMD applies the coding scheme from the training phase to a larger number of annual reports. In order to improve runtime, AIMD automatically pre-processes the texts to be analysed, eliminating all stopwords and putting words into a canonical morphological form as described in Appendix C. This means that we only need to consider permutations of elements of standardised *N*-grams: applying the coding scheme is equivalent to searching for standardised *N*-grams within a text regardless of their word order.

The final step is to count the number of times the standardised *N*-grams occur in each annual report. If we accept that the meaning of all *N*-grams relating to a particular disclosure dimension is identical (see Section 3.2), we can measure the intensity of disclosure in a text simply by using category counts (Weber, 1990, pp. 56–58), that is, measuring the frequency of items relating to this disclosure dimension. Both the number of marked *N*-grams and the number of marked words are candidates for the category counts. AIMD uses word-based category counts, as marked *N*-grams can overlap within a text; word-based category counts are also easier to implement.

Figure 2 summarises the steps involved in the application phase. I automated the entire application phase using proprietary software written in C#; no human interaction is thus required.

## 4. Reliability and Validity of AIMD

In human-scored schemes, reliability – that is, the reproducibility of the measurement – is a major concern (Marston and Shrives, 1991, p. 197; Healy and Palepu, 2001, p. 427). By contrast, in the case of software algorithms with no random components, reliability is perfect (Weber, 1990, p. 41; Krippendorff, 2004,

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**Table 2**. Marginal contribution of annual reports in creating the coding scheme

| | Disclosure dimension | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| # report | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) | Total |
| 1 | 844 | 515 | 565 | 351 | 258 | 318 | 277 | 367 | 306 | 223 | 4024 |
| | *39.1%* | *25.7%* | *29.2%* | *23.3%* | *29.4%* | *30.5%* | *44.1%* | *33.8%* | *28.1%* | *22.7%* | *30.2%* |
| 2 | 334 | 278 | 228 | 199 | 147 | 146 | 99 | 156 | 148 | 134 | 1869 |
| | *15.5%* | *13.9%* | *11.8%* | *13.2%* | *16.7%* | *14.0%* | *15.8%* | *14.4%* | *13.6%* | *13.6%* | *14.0%* |
| 3 | 190 | 185 | 163 | 139 | 95 | 85 | 52 | 106 | 95 | 105 | 1215 |
| | *8.8%* | *9.2%* | *8.4%* | *9.2%* | *10.8%* | *8.2%* | *8.3%* | *9.8%* | *8.7%* | *10.7%* | *9.1%* |
| … | … | … | … | … | … | … | … | … | … | … | … |
| 18 | 15 | 21 | 22 | 19 | 6 | 12 | 3 | 9 | 13 | 12 | 132 |
| | *0.7%* | *1.0%* | *1.1%* | *1.3%* | *0.7%* | *1.2%* | *0.5%* | *0.8%* | *1.2%* | *1.2%* | *1.0%* |
| … | … | … | … | … | … | … | … | … | … | … | … |
| 22 | 9 | 12 | 14 | 13 | 2 | 7 | 1 | 4 | 7 | 6 | 75 |
| | *0.4%* | *0.6%* | *0.7%* | *0.9%* | *0.2%* | *0.7%* | *0.2%* | *0.4%* | *0.6%* | *0.6%* | *0.6%* |
| 23 | 9 | 11 | 14 | 12 | 1 | 6 | 0 | 3 | 5 | 5 | 66 |
| | *0.4%* | *0.5%* | *0.7%* | *0.8%* | *0.1%* | *0.6%* | *0.0%* | *0.3%* | *0.5%* | *0.5%* | *0.5%* |
| 24 | 4 | 9 | 9 | 9 | 0 | 5 | 0 | 2 | 5 | 4 | 47 |
| | *0.2%* | *0.4%* | *0.5%* | *0.6%* | *0.0%* | *0.5%* | *0.0%* | *0.2%* | *0.5%* | *0.4%* | *0.4%* |
| Total | 2157 | 2007 | 1935 | 1508 | 879 | 1042 | 628 | 1085 | 1090 | 983 | 13,314 |
| | *100%* | *100%* | *100%* | *100%* | *100%* | *100%* | *100%* | *100%* | *100%* | *100%* | *100%* |

The table shows the contribution of annual reports used in creating the coding scheme to the standardised *N*-gram lists for specific disclosure dimensions. A key to disclosure dimensions 1–10 is given in Table 1. Non-italic figures indicate the absolute contribution in terms of the number of standardised *N*-grams; italic figures indicate the relative contribution as a share of the total for all 24 reports.

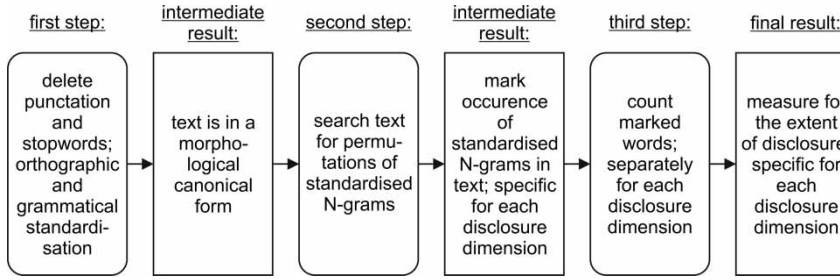



**Figure 2.** Application phase: content analysis using the coding scheme

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p. 259). The application phase of AIMD is thus perfect in terms of reliability, being entirely objective. In the training phase a certain subjectivity remains as the *N*-grams are extracted manually. Given the considerable manual effort involved in extracting lists of *N*-grams from the reports, I do not address the reliability of the training phase with a test-retest design but by the validity of the entire measurement process as discussed below.

The question of validity concerns whether the measurement procedure really measures what it claims to do. Three types of validity exist: content validity, criterion-related validity and construct validity (Davis, 1974, pp. 33–35). Content validity is the weakest type of validity (Weber, 1990, p. 18) and is difficult to quantify. I present evidence for the content validity of AIMD in Appendices D and E. In Section 4.1, below, I evaluate whether AIMD has criterion validity, that is, if a sound theoretically predicted relation can be confirmed empirically using the measurements of disclosure produced by AIMD. In Section 4.2 I examine whether AIMD has construct validity, that is, if it converges with other popular techniques for measuring corporate disclosure.

### *4.1. Criterion Validity Tests*

The mainstream literature hypothesises a negative relationship between corporate disclosure and information asymmetry between market participants (e.g. Verrecchia, 1982; Diamond, 1985; Baiman and Verrecchia, 1996; Healy *et al.*, 1999; Brockman and Chung, 2001; Easley and O'Hara, 2004; Heflin *et al.*, 2005; Chen *et al.*, 2007). Following Jayaraman (2008), the criterion validity tests which I use here rely on the 'bid-ask spread' as a measure of information asymmetry in a market-maker environment, and the probability of information-based trading (PIN) (Easley *et al.*, 1996, 1997, 2010) as a measure of the level of informed trading, that is, available private information. Limited time series are available for these microstructure measures, but Amihud (2002, p. 32) argues that they are superior to other liquidity measures for which a larger number of time series are available.

Looking at common control variables (e.g. Chiang and Venkatesh, 1988; Leuz, 2003; Agrawal *et al.*, 2004; Jayaraman, 2008), the following regression addresses the spread reduction effect (variables are defined in Table 3):

$$\begin{aligned}\text{SPREAD}=&\alpha+\beta_1\text{AIMD}+\beta_2\text{SIZE}+\beta_3\text{PROFIT}+\beta_4\text{PROFIT_VOL}+\beta_5\text{MB}\\&+\beta_6\text{LEVERAGE}+\beta_7\text{SH_TURN}+\beta_{8\ldots16}\text{INDUSTRY}+\beta_{17\ldots19}\text{YEAR}+\varepsilon.\end{aligned} \tag{1}$$

An equivalent regression addresses the reduction of the amount of private information due to disclosure (variables are defined in Table 4):

$$\begin{aligned}\text{PIN}=&\ \alpha\ +\ \beta_1\text{AIMD}+\ \beta_2\text{SIZE}+\ \beta_3\text{PROFIT}+\beta_4\text{PROFIT\ _VOL}+\beta_5\text{MB}\\&+\beta_6\text{LEVERAGE}+\beta_7\text{SH\ _TURN}+\beta_{8\ldots16}\text{INDUSTRY}+\beta_{17\ldots24}\text{YEAR}+\ \varepsilon.\end{aligned} \tag{2}$$

Table 3 gives the results of the multivariate spread regressions based on 13,710 firm-year observations between 2005 and 2008 for firms regulated by the SEC. Panel A lists the coefficients, *t*-statistics and other information for models with AIMD, proxied by the factor extracted from the AIMD *N*-gram frequencies for all 10 disclosure dimensions. The results for Model 1 in Panel A are OLS estimates. As the Breusch-Pagan statistics indicate potential heteroscedasticity in errors, Model 2 gives estimates with heteroscedasticity-consistent standard errors (White, 1980) which are robust to within cluster correlation (Rogers, 1993), that is, clustered or 'Rogers' standard errors using firm-level clusters. Because of the high number of firm-level clusters (4775) relative to the sample size (13,710), Model 3 gives the results of an estimation with Rogers standard errors for two-way YEAR and INDUSTRY clusters according to Cameron *et al.* (2006). I address potential nonlinear relations by rank regressions; Model 4 is the equivalent rank regression of Model 2, and Model 5 is the equivalent rank regression of Model 3. Panel B gives the results for individual AIMD disclosure dimensions. In all other respects the models in Panel B are equivalent to Model 3 of Panel A.

Low 'variance inflation factors' (VIFs, not reported) reveal no multicollinearity problems. The significant negative estimated coefficients for the disclosure variables indicate that AIMD has explanatory power for the bid-ask spread, which decreases as the measurement of disclosure increases. The criterion validity holds for both the factored total measurement of disclosure and each individual dimension of disclosure. The results are robust when subjected to additional controls (total share returns and share volatility, not reported) and different variable specifications (not reported), including Stoll's proportional quoted bid-ask spread (2000, p. 1480) as an alternative SPREAD proxy, the unweighted sum of the AIMD *N*-gram frequencies of the 10 disclosure

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dimensions as an alternative aggregated AIMD proxy, and total assets and total employees as alternative SIZE proxies.

Similarly, Table 4 reports the results of the multivariate PIN regressions on AIMD total disclosure scores based on 11,606 firm-year observations between 1994 and 2001 of firms regulated by the SEC. Panel A gives the results for models with total disclosure measured by the factored *N*-gram frequency over all 10 disclosure dimensions. Again the results for Model 1 are OLS estimates, the results for Model 2 are Rogers corrected with firm-level clusters, and the results for Model 3 are Rogers corrected using two-way clusters for YEAR and INDUSTRY. Model 4 is a rank regression version of Model 2, and Model 5 a rank regression version of Model 3. Panel B gives the results for individual disclosure dimensions based on Panel A's Model 3 specifications.

All VIFs (not reported) are low, providing no indication for multicollinearity. The estimated coefficients are negative and significant for both the total AIMD measure and the AIMD measures of each individual disclosure dimension. The results are robust when subjected to the additional controls and alternative variable specifications as discussed for the SPREAD regression, except for the fact that no alternative PIN proxy is available. Accordingly, the PIN decreases in line with increasing disclosure levels as measured by AIMD, and criterion validity also holds for the PIN regressions.

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Table 3. Multivariate regression of bid–ask spread on disclosure

| Panel A: Total disclosure regressions | | | | | |
|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 |
| AIMD | −0.12*** | −0.12*** | −0.12** | −0.04*** | −0.04*** |
| | (−15.36) | (−11.11) | (−3.46) | (−6.11) | (−3.86) |
| SIZE | −0.37*** | −0.37*** | −0.37*** | −0.29*** | −0.29*** |
| | (−36.53) | (−16.27) | (−4.86) | (−23.54) | (−8.47) |
| PROFIT | −0.03*** | −0.03 | −0.03 | −0.12*** | −0.12*** |
| | (−3.97) | (−1.94) | (−1.06) | (−17.54) | (−8.05) |
| PROFIT_VOL | 0.18*** | 0.18*** | 0.18*** | 0.21*** | 0.21*** |
| | (20.79) | (8.60) | (8.75) | (28.75) | (19.15) |
| MB | −0.14*** | −0.14*** | −0.14*** | −0.11*** | −0.11*** |
| | (−21.59) | (−11.16) | (−6.52) | (−17.92) | (−17.34) |
| LEVERAGE | 0.10*** | 0.10*** | 0.10** | 0.08*** | 0.08*** |
| | (13.69) | (8.82) | (3.44) | (11.59) | (7.42) |
| SH_TURN | 0.06*** | 0.06*** | 0.06*** | −0.48*** | −0.48*** |
| | (8.64) | (7.12) | (3.76) | (−45.98) | (−17.15) |
| INDUSTRY | Yes | Yes | Yes | Yes | Yes |
| YEAR | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.34 | 0.34 | 0.34 | 0.791 | 0.791 |
| *N* | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 |
| Breusch–Pagan | 13,099*** | | | | |

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Panel B: Regressions for specific disclosure dimensions

| | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) |
|---|---|---|---|---|---|---|---|---|---|---|
| AIMD | −0.12*** | −0.11** | −0.11** | −0.09** | −0.24*** | −0.11** | −0.08*** | −0.10*** | −0.07** | −0.08** |
| | (−3.73) | (−2.95) | (−3.14) | (−2.98) | (−4.88) | (−3.49) | (−3.78) | (−4.39) | (−3.00) | (−3.00) |
| $R^2$ | 0.339 | 0.338 | 0.339 | 0.335 | 0.366 | 0.338 | 0.334 | 0.336 | 0.333 | 0.334 |
| $N$ | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 | 13,710 |
| Controls (not reported): SIZE, PROFIT, PROFIT_VOL, MB, LEVERAGE, SH_TURN, INDUSTRY, YEAR | | | | | | | | | | |

The dependent variable is the bid–ask spread (SPREAD) as a yearly average of daily differences in ask and bid prices per the average of bid and ask price. In Panel A, AIMD is the factor (KMO 0.877; one eigenvalue of 7.15 above one) extracted from AIMD measures of the 10 disclosure dimensions based on the whole text of the 10-K, 10-K405 and 10KSB, respectively. In Panel B, AIMD is the disclosure dimension-specific AIMD. A key to disclosure dimensions 1–10 is given in Table 1. SIZE is the natural logarithm of total sales. PROFIT is the return on assets. PROFIT_VOL is the standard deviation of profitability over the last three years. MB is the market-to-book ratio. LEVERAGE is the debt–equity ratio. SH_TURN is average of daily share turnover over the last year. YEAR represents four-year dummies for 2005–2008. INDUSTRY represents 10 industry dummies based on Standard Industrial Codes (SIC) aggregated using Kenneth R. French's methodology. A list of the SICs making up the industry portfolios is available at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

All variables are measured at the filing date of the 10-K, 10-K405 and 10KSB, respectively. All metric variables are standardised and winsorised following Bushee and Noe (2000, p. 182), Oswald and Zarowin (2007, p. 724), Bushee and Goodman (2007, p. 302) and Lee *et al.* (2008, p. 734) at the 0.02 level at both extremes to adjust for potential outliers. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

The table reports regression coefficients. *T*-statistics are given in parentheses. In Panel A, Model 1 reports the results of an OLS regression. Model 2 lists the results of a regression with White standard errors which are robust to within cluster correlation (i.e. Rogers standard errors) using clusters for individual companies. Model 3 is similar to Model 2, but uses two-way clusters for YEAR and INDUSTRY following Cameron *et al.* (2006). Model 4 reports the results of a rank regression with Rogers standard errors using clusters for individual companies. Model 5 is similar to Model 4 but uses clusters for both YEAR and INDUSTRY. In Panel B all models report the results of a regression with Rogers standard errors using clusters for both YEAR and INDUSTRY.

### *4.2. Construct Validity Tests*

If AIMD has construct validity, it should converge with other techniques for measuring disclosure discussed in the literature (see Section 2). The first test focuses on the rating of the AIMR (Lang and Lundholm, 1996; Sengupta, 1998; Bushee and Noe, 2000; Botosan and Plumlee, 2002). This receiver rating covers around 400–500 firms each year between 1978 and 1996. Due to the limited availability of earlier annual reports, I only consider the 1996 rating. AIMR rating criteria vary for different industries so it was necessary to normalise ratings for the sake of comparisons. As an alternative to simple intra-industry rankings, Botosan and Plumlee (2002, p. 28) suggest taking the percentage deviation of a company's rating from the industry average. The analysis is based on Spearman rank correlations[2] because of the unknown linearity of the relation between the AIMR rating and AIMD category counts. Panel 1 of Table 5 reports the correlation coefficients for each test.

The AIMR ranks correlate significantly with AIMD for all disclosure dimensions except *governance information*, where the correlation is insignificant by a small margin. The AIMR ranks also correlate with total disclosure as proxied by the factor extracted from the AIMD measurements for all disclosure dimensions. The results for the AIMR percentage deviation from the industry average are similar. For the total disclosure level and all disclosure dimensions

Table 4. Multivariate regression of the probability of information-based trading (PIN) on disclosure

| Panel A: Total disclousure regressions | | | | | |
|---|---|---|---|---|---|
| | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 |
| AIMD | −0.06*** | −0.06*** | −0.06*** | −0.03** | −0.03** |
| | (−7.88) | (−6.22) | (−3.56) | (−2.97) | (−2.92) |
| SIZE | −0.70*** | −0.70*** | −0.70*** | −0.27*** | −0.27*** |
| | (−70.44) | (−41.89) | (−14.33) | (−15.45) | (−13.89) |
| PROFIT | 0.02 | 0.02 | 0.02 | 0.01 | 0.01 |
| | (1.87) | (1.45) | (1.34) | (1.66) | (0.97) |
| PROFIT_VOL | −0.05*** | −0.05*** | −0.05** | 0.03*** | 0.03*** |
| | (−6.43) | (−4.60) | (−3.28) | (3.83) | (4.27) |
| MB | −0.22*** | −0.22*** | −0.22*** | −0.15*** | −0.15*** |
| | (−28.60) | (−18.92) | (−10.78) | (−14.97) | (−11.15) |
| LEVERAGE | 0.15*** | 0.15*** | 0.15*** | 0.07*** | 0.07*** |
| | (19.37) | (10.70) | (4.72) | (7.24) | (5.77) |
| SH_TURN | 0.01 | 0.01 | 0.01 | −0.52*** | −0.52*** |
| | (1.19) | (0.94) | (0.45) | (−28.05) | (−11.22) |
| INDUSTRY | Yes | Yes | Yes | Yes | Yes |
| YEAR | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.502 | 0.502 | 0.502 | 0.638 | 0.638 |
| *N* | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 |
| Breusch–Pagan | 1634.38*** | | | | |

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Panel B: Regressions for specific disclosure dimensions

| | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) |
|---|---|---|---|---|---|---|---|---|---|---|
| AIMD | −0.06** | −0.06*** | −0.06*** | −0.05*** | −0.10*** | −0.07*** | −0.06*** | −0.06** | −0.04** | −0.04* |
| | (−2.80) | (−3.48) | (−3.59) | (−4.93) | (−4.56) | (−4.02) | (−4.46) | (−3.21) | (−2.96) | (−2.40) |
| $R^2$ | 0.501 | 0.502 | 0.501 | 0.501 | 0.505 | 0.503 | 0.501 | 0.502 | 0.500 | 0.500 |
| $N$ | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 | 11,606 |
| Controls (not reported): SIZE, PROFIT, PROFIT_VOL, MB, LEVERAGE, SH_TURN, INDUSTRY, YEAR | | | | | | | | | | |

The dependent variable is the probability of information-based trading (PIN) as measured by Easley *et al.* (2010). Raw data is taken from http://sites.google.com/site/hvidkjaer/data. In Panel A, AIMD is the factor (KMO 0.889; one eigenvalue of 7.47 above one) extracted from AIMD measures of the 10 disclosure dimensions based on the whole text of the 10-K, 10-K405 and 10KSB, respectively. In Panel B, AIMD is the disclosure dimension-specific AIMD. A key to disclosure dimensions 1–10 is given in Table 1.

SIZE is the natural logarithm of total sales. PROFIT is the return on assets. PROFIT_VOL is the standard deviation of profitability over the last three years. MB is the market-to-book ratio. LEVERAGE is the debt–equity ratio. SH_TURN is average of daily share turnover over the last year. YEAR represents eight-year dummies for 1994–2001. INDUSTRY represents 10 industry dummies based on Standard Industrial Codes (SIC) aggregated using Kenneth R. French's methodology. A list of the SICs which form the industry portfolios is available at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

All variables are measured at the year end of the respective PIN datapoint. All metric variables are standardised and winsorised following Bushee and Noe (2000, p. 182), Oswald and Zarowin (2007, p. 724), Bushee and Goodman (2007, p. 302) and Lee *et al.* (2008, p. 734) at the 0.02 level at both extremes to adjust for potential outliers. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

The table shows regression coefficients. *T*-statistics are given in parentheses. In Panel A, Model 1 reports the results of an OLS regression. Model 2 lists the results of a regression with White standard errors which are robust to within cluster correlation (i.e. Rogers standard errors) using clusters for individual companies. Model 3 is similar to Model 2 but uses two-way clusters for YEAR and INDUSTRY following Cameron *et al.* (2006). Model 4 reports the results of a rank regression with Rogers standard errors using clusters for individual companies. Model 5 is similar to Model 4 but uses clusters for both YEAR and INDUSTRY. In Panel B all models report the results of a regression with Rogers standard errors using clusters for both YEAR and INDUSTRY.

except *research and development information*, the correlation is significantly positive. Overall there is thus strong empirical evidence that AIMD has construct validity when compared to the AIMR rating.

I also test construct validity by comparing AIMD to three observer approaches (*Standard & Poor's (S&P) Transparency and Disclosure Rating* and two disclosure ratings published by business magazines). The S&P rating is based on a checklist which covers items in the three categories (1) ownership structure and investor rights, (2) financial transparency and information disclosure, and (3) board structure and process (Patel and Dallas, 2002). Accordingly it only addresses the disclosure dimensions *financial information*, *corporate governance*, *capital market*, *corporate strategy* and *corporate environment*. Patel and Dallas (2002) provide raw data from the S&P ranking for the 10-Ks of S&P 500 firms for the year 2002. Panel 2 of Table 5 gives the results of the correlation analysis of this S&P composite ranking and the factored AIMD of the 10-Ks in question. The positive and significant coefficients for the relevant disclosure dimensions indicate that AIMD has construct validity when measured against the S&P ranking.

The German magazine *Managermagazin* publishes comprehensive manual disclosure scorings produced by the University of Münster[3] for large German companies. My construct validity tests compare AIMD with content ratings from *Managermagazin* for 2003–2006 (Papendick, 2004; Döhle, 2005, 2006, 2007). Panel 3 of Table 5 gives the resulting Spearman correlation coefficients: they are positive and significant for all disclosure dimensions, indicating that AIMD has construct validity when compared to the *Managermagazin* disclosure rating. A similar analysis using the ratings published by the magazines *Capital* and *Focus Money* (see Daske and Gebhardt, 2006) has similar results (not reported), supporting the construct validity of AIMD. The rating produced by the *Center for International Financial Analysis & Research (CIFAR)* (CIFAR, 1995; Hope, 2003, pp. 266–269; Baumann and Nier, 2004, pp. 33–35) was discontinued in 1993 (Khanna *et al.*, 2004, p. 483) and could not be used as a benchmark due to the limited availability of annual reports.

Finally, I tested construct validity against a sender approach based on companies' own assessments of their disclosure in 2005. In 2006 I collected raw data from all 600 firms listed on the German stock exchange using a questionnaire. A total of 153 firms (25.5%) responded, of which 103 (17.2%) returned a completed questionnaire. Self-assessment by companies may differ from other benchmarks, as respondents may try to paint an overly positive picture of their level of disclosure. The positive and significant Spearman correlation coefficients reported in Panel 4 of Table 5 support the construct validity of AIMD.

To summarise, the results show that AIMD displays perfect reliability, has criterion validity in the information asymmetry reduction effect and has construct validity when tested against several other techniques for measuring disclosure. This validity holds for several years and across different countries (i.e. different

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**Table 5**. Correlation between AIMD and other techniques for measuring disclosure

| | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) | Total | *N* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Panel 1: AIMR ranking | | | | | | | | | | | | |
| (A) | −0.31** | −0.16* | −0.30** | −0.16* | −0.20** | −0.29** | −0.12 | −0.22** | −0.22** | −0.36** | −0.27** | 169 |
| | (0.000) | (0.018) | (0.000) | (0.021) | (0.005) | (0.000) | (0.063) | (0.002) | (0.002) | (0.000) | (0.000) | |
| (B) | 0.218** | 0.102 | 0.180** | 0.137* | 0.014 | 0.226** | 0.171* | 0.187** | 0.133* | 0.211** | 0.184** | 169 |
| | (0.002) | (0.094) | (0.010) | (0.038) | (0.427) | (0.002) | (0.013) | (0.007) | (0.042) | (0.003) | (0.008) | |
| Panel 2: Standard & Poor's Transparency & Disclosure Rating | | | | | | | | | | | | |
| | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) | Total | *N* |
| Cat. 1 | 0.054 | −0.028 | 0.072 | 0.051 | −0.089 | 0.025 | 0.177** | 0.130** | 0.027 | 0.083 | 0.068 | 437 |
| | (0.129) | (0.279) | (0.068) | (0.142) | (0.031) | (0.298) | (0.000) | (0.003) | (0.290) | (0.041) | (0.079) | |
| Cat. 2 | 0.225** | 0.125 | 0.211 | 0.214 | −0.082 | 0.235** | 0.226** | 0.227** | 0.214** | 0.264 | 0.212 | 437 |
| | (0.000) | (0.004) | (0.000) | (0.000) | (0.044) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |
| Cat. 3 | 0.211 | 0.103 | 0.135 | 0.218 | −0.048 | 0.197 | 0.210** | 0.194 | 0.174 | 0.199 | 0.173 | 437 |
| | (0.000) | (0.06) | (0.002) | (0.000) | (0.158) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |
| Total | 0.170** | 0.060 | 0.158 | 0.182 | −0.113 | 0.187** | 0.245** | 0.212** | 0.159** | 0.212 | 0.171** | 437 |
| | (0.000) | (0.104) | (0.001) | (0.000) | (0.009) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |
| Panel 3: *Managermagazin* scoring | | | | | | | | | | | | |
| | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) | Total | *N* |
| 2006 | 0.607** | 0.604** | 0.536** | 0.628** | 0.340** | 0.577** | 0.511** | 0.541** | 0.643** | 0.482** | 0.605** | 101 |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |
| 2005 | 0.620** | 0.631** | 0.591** | 0.674** | 0.341** | 0.628** | 0.599** | 0.571** | 0.665** | 0.549** | 0.660** | 108 |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |
| 2004 | 0.587** | 0.613** | 0.607** | 0.531** | 0.245** | 0.583** | 0.524** | 0.532** | 0.620** | 0.501** | 0.604** | 89 |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.010) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |
| 2003 | 0.647** | 0.616** | 0.643** | 0.586** | 0.346** | 0.623** | 0.541** | 0.598** | 0.711** | 0.584** | 0.651** | 87 |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.001) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | |

(*Continued*)

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**Table 5.** Continued

Panel 4: Companies' own evaluation of their disclosure

| | (01) | (02) | (03) | (04) | (05) | (06) | (07) | (08) | (09) | (10) | Total | *N* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2005 | 0.266** (0.007) | 0.369** (0.000) | 0.322** (0.001) | 0.393** (0.000) | 0.479** (0.000) | 0.318** (0.002) | 0.297** (0.003) | 0.255** (0.009) | 0.329** (0.001) | 0.317** (0.002) | 0.381** (0.000) | 85 |

The table shows Spearman correlation coefficients for specific disclosure dimensions with the respective construct validity benchmark and, in parentheses, the coefficients' significances. A key to disclosure dimensions 1–10 is given in Table 1. AIMD total disclosure is the factor extracted from AIMDs of the 10 disclosure dimensions. $^{*}p < 0.05$, $^{**}p < 0.01$.

In Panel 1, Model (A) lists the results for the AIMR inter-industry ranking and Model (B) the results for the relative deviation of AIMR scores from the industry average.

In Panel 2, the *ownership structure and investor rights* category (cat. 1) focuses on 'transparency and concentration of ownership', that is, information relating to the capital market, as well as 'voting and shareholder meeting procedures', that is, a mixture of information relating to corporate governance and the capital markets. The *financial transparency and information* disclosure category (cat. 2) subsumes 'business focus' information relating to the corporate strategy and the corporate environment disclosure dimension, and 'accounting policy review' and 'accounting policy details' addressed in the financial disclosure dimension. This category also covers 'related party structure and transaction', that is, information relating to the capital market and corporate governance, and 'information on auditors', that is, information about corporate governance. The *board structure and processes category* (cat. 3) focuses exclusively on corporate governance information (quotations are taken from Patel and Dallas, 2002, pp. 19–21). The S&P ranking does not cover information about the sales market and customers, employees, research and development, social and environmental responsibility, and the value chain.

legislative systems). Overall AIMD may therefore be considered a valid and reliable approach for measuring levels of corporate disclosure.

## 5. Conclusions

Empirical research on voluntary disclosure lacks an appropriate measurement technique for evaluating disclosure. The disclosure ratings produced by financial organisations and human-scored, self-constructed indices suffer from a number of disadvantages including limited availability of data, questionable reliability and often dubious validity. A further problem is the extreme labour-intensity of human-scored indices and the lack of an up-to-date set of data which could form the basis for large-scale empirical research.

In this paper I introduce AIMD, a computerised measure of corporate disclosure using artificial intelligence. AIMD measures disclosure along 10 information dimensions by analysing annual reports published in English. It uses an information-retrieval vector space model based on AI techniques employing an abstract world model. In the training phase, a coding scheme is created. This then forms the basis for the application phase, in which disclosure is measured on the basis of texts published by the companies.

The application phase does not require human involvement and is perfectly reliable. The extensive validity tests reported in this paper demonstrate that AIMD has both criterion and construct validity. Multivariate regression analyses based on large samples show that AIMD is significantly negatively associated with information asymmetry. Construct validity tests show that it is also significantly associated with other popular techniques for measuring disclosure, including the AIMR and S&P ratings, human-scored approaches and assessment by companies of their own disclosure levels.

AIMD can measure disclosure on the basis of different kinds of texts, including annual or quarterly reports, the minutes of meetings with analysts, press releases, records of conference calls and company websites. Its applicability for large-scale empirical research is demonstrated using a sample of 127,895 firm-year observations (see Appendix D). Appendix E discusses how AIMD can also be used to examine the distribution of disclosure *within* individual annual reports.

AIMD is a reliable and cost-effective procedure for measuring the level of disclosure by companies. It does not suffer from the disadvantages of existing techniques for measuring disclosure (e.g. Marston and Shrives, 1991; Beattie and Thomson, 2007). It thus makes it possible to re-examine the findings of earlier research using larger sample sizes, opening up new avenues for empirical research in the area of disclosure. However, future research should address the question of whether AIMD's validity varies between industries and is stable over time. AIMD may also be enhanced by considering additional information dimensions to distinguish historical vs. forward-looking information, financial vs. non-financial and quantitative vs. qualitative information.

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**Data Availability Statement**

AIMD data is available from the author on request.

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## Appendix A: Selecting Textual Documents for Creating the Coding Scheme

This appendix documents the selection procedure for the 24 annual reports used in the training phase as a basis for extracting the *N*-gram lists described in Appendix B.

The *N*-gram lists should be comprehensive, allowing AIMD to map all the potential *N*-grams of a text which are relevant for disclosure. Accordingly the reports used for creating the coding scheme should reflect the full diversity of corporate disclosure. At the same time the scheme should not be excessively complex as this would have a negative impact on its performance in the application phase.

First of all I chose annual reports to be the most appropriate type of textual documents as they are the most comprehensive and important instrument of corporate disclosure (e.g. Courtis, 1986, p. 1; Botosan, 1997, p. 331; White and Hanson, 2002, p. 291; Guthrie *et al.*, 2004, p. 283; Ho and Wong, 2004). Furthermore, annual reports are readily available from company websites, the websites of stock exchanges, and other websites as well as online databases.

Second, it is necessary to select the specific annual reports to be used. In order for the coding scheme to be complete, the reports chosen should cover the full range of corporate disclosure comprehensively. I argue below why 24 annual reports from 2003 published by the German DAX-30 firms in English are an appropriate set of reports to train AIMD. However, I recognise that the selection procedure still contains a certain amount of subjectivity.

Empirical evidence suggests that German annual reports contain an exceptionally high level of disclosure in the non-financial disclosure dimensions (Roberts, 1990; Meek *et al.*, 1995, p. 564; Adams *et al.*, 1995, 1998; Adams and Kuasirikun, 2000), that is, 9 of the 10 disclosure dimensions. For example, Gray *et al.* (1996, p. 171) find that 'German companies are leading mainland Europe in disclosing various types of environmental information'. On the other hand, many researchers claim that their level of financial disclosure is low (Alford *et al.*, 1993; Harris *et al.*, 1994; Jaggi and Low, 2000; Hope, 2003, pp. 251–254; Hail and Leuz, 2006, pp. 494–495). This claim must be re-examined in the

light of recent developments in financial accounting and reporting regulations in Germany. Since 1 January 2003 all firms listed in the prime segment of the German stock exchange ('Prime Standard') have been obliged to use International Accounting Standards (IAS/IFRS).[4] Empirical evidence suggests that there is no difference in the quality of US GAAP and IAS/IFRS annual accounts (Alford and Jones, 1998; Leuz, 2003). I therefore conclude that German Prime Standard annual reports provide a level of financial disclosure similar to that found internationally.

It is widely accepted that cross-listing increases the disclosure level of companies (Cooke, 1989, 1991, 1993; Meek *et al.*, 1995; Ahmed and Courtis, 1999; Archambault and Archambault, 2003; Krishnamurti *et al.*, 2005; Clarkson *et al.*, 2006; Grüning, 2007). Most of the 24 companies which I examine are cross-listed on up to six major capital markets and therefore subject to a variety of foreign disclosure requirements. Two-thirds of the companies are cross-listed in the US and subject to regulation by the SEC. Many are cross-listed in Switzerland, the UK and other European markets. Some companies are also cross-listed in the Far East.

Company size is another important determinant of disclosure (Ahmed and Courtis, 1999; Francis *et al.*, 2005; Chavent *et al.*, 2006). The 24 annual reports which I examine were published by very large, highly cross-listed firms. Therefore, empirical evidence suggests that the overall level of corporate disclosure is high for these German Prime Standard annual reports.

Table A1 lists the 24 DAX-30 companies whose 2003 annual reports I use as a basis for extracting the *N*-gram lists. I specifically excluded financial institutions and insurance companies from the sample.[5] In Section 3.3 I examine whether the relatively low number of firms is sufficient to provide a comprehensive coding scheme.

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## Appendix B: Extracting *N*-Gram Lists

For each of the 10 disclosure dimensions a separate list of *N*-grams is manually created by mapping all the *N*-grams appearing in the 24 annual reports onto either a disclosure dimension or a residual category of non-disclosure content. All the syntactic characteristics of *N*-grams in the texts are considered in the form in which they appear, but duplicates of *N*-grams are ignored. For example, the first occurrence of 'sales organisation' in one of the texts constitutes an *N*-

**Table A1**. Companies whose annual reports were used to create the coding scheme

| | | | |
|---|---|---|---|
| Adidas-Salomon AG | DaimlerChrysler AG | Henkel KGaA | SAP AG |
| Altana AG | Deutsche Lufthansa AG | Infineon Technologies AG | Schering AG |
| BASF AG | Deutsche Post AG | Linde AG | Siemens AG |
| Bayer AG | Deutsche Telekom AG | MAN AG | ThyssenKrupp AG |
| BMW AG | E.ON AG | Metro AG | TUI AG |
| Continental AG | Fresenius AG | RWE AG | Volkswagen AG |

gram list entry for the *sales market and customers* disclosure dimension. If this *N*-gram appears again in the same report or another report, no additional entry is made in the *N*-gram list, but a different *N*-gram list entry results for the plural 'sales organisations'. The *N*-gram lists do not consider punctuation and they are not case sensitive except for abbreviations.[6] They are sensitive to spelling, however, so British and American variants of *N*-grams are recorded as two separate entries, for example. Obvious misspellings are corrected. The *N*-gram lists are also sensitive to inflection, so each conjugated or declined form (including comparatives) constitutes a different entry in the *N*-gram lists.

*N*-grams are selected on the basis of the 'connotative categorical equivalent' criterion. Thus if a monogram clearly relates to a disclosure dimension, it constitutes a list entry; if not, it is mapped to the residual category. If subsequent words allow a mapping to a disclosure dimension, the resulting bigram, trigram (and so on) forms a list entry. For example, the monogram 'number' is mapped to the residual category as it does not clearly relate to a disclosure dimension, but the *N*-gram 'number of employees' clearly relates to the *employees* disclosure dimension and so constitutes an *N*-gram list entry for this dimension.

Following North *et al.* (1963, pp. 134–135) and Kothari *et al.* (2009), the *N*-gram lists are not mutually exclusive. About 8.8% (1178) of the standardised *N*-grams do not relate to a single content category. For example, 'pension accruals' relates to both the *employees* and the *financial disclosure* dimension. Table 1 lists some examples of *N*-grams for each disclosure dimension. The manual extraction from the 24 annual reports provides a total of 34,819 different *N*-grams (from a total of 5096 norm pages of 1800 characters each).

## Appendix C: Standardising *N*-Gram Lists

To reduce complexity and increase ease of use, it is necessary to normalise the *N*-gram lists. The aim here is to make sure that the coding scheme is independent of (1) meaningless 'stopwords', (2) the morphology of different elements within *N*-grams, and (3) permutations of the different elements within *N*-grams. For example, the standardised bigram 'employee number' should also represent the tetragram 'number of the employees'.

'Stopwords' are words which do not contribute to the content of a text. They include articles, pronouns, conjunctions, prepositions, generic verbs and such like. All such words are deleted from the *N*-grams. For example, the *N*-gram 'number of the employees' reduces to 'number employees'. Here I use the list of 319 stopwords published by the Information Retrieval Group of the University of Glasgow.

Next all the different elements within *N*-grams are put into their canonical morphologic form. This means normalising both their grammar and their spelling. AIMD uses what is known as 'dictionary-based stemming' (Kowalski, 1997, pp. 71–72) based on the Automatically Generated Inflection Database (AGID), which gives the 112,503 roots of 281,904 inflected forms. For example, 'organisations' and 'organisation's' are reduced to 'organisation' (British spelling

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variant) and 'organizations' and 'organisation's' are reduced to 'organisation' (American spelling variant).

Next the spelling of items is standardised to account for British, American and Canadian variants, for example. For this purpose I use the Variant Conversion Info (VarCon), which gives spelling variants for 16,019 words.

Standardising the permutations of *N*-grams involves removing non-semantic differences based on the order in which the individual elements within an *N*-gram appear. Different grammatical structures in English result in different word orders within sentences (Murphy, 2007). When measuring corporate disclosure, only semantic differences are of importance. For instance, there is no relevant semantic difference between the *N*-grams 'flexible arrangements of work time' and 'an arrangement of flexible working time': apart from the stopwords and morphology, the only thing that differs is the word order. *N*-grams are therefore standardised for such permutations.

Figure C1 summarises the standardisation process and gives some examples. As a result of the standardisation, the total number of items on the *N*-gram lists falls from 34,819 to 13,314.

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## Appendix D: Disclosure by Companies Regulated by the SEC

Figure D1 shows the average AIMD category counts by year and disclosure dimension for all 127,895 10-K, 10-K405 and 20-F that report about financial years between 1996 and 2008 and are filed electronically with the SEC until the end of the second quarter 2010. For the sake of clarity, the figure only shows the development over time of four dimensions: *financial information* and *capital markets* (the two most important dimensions), *research and development* (the least important dimension) and *employees* (somewhere in between).

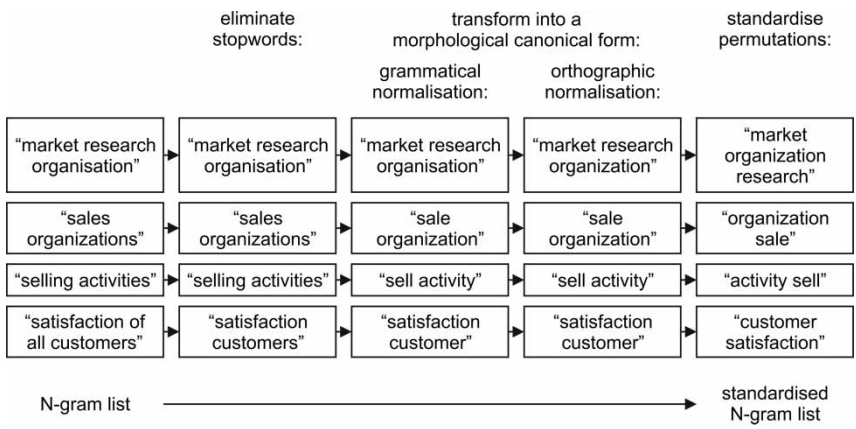



**Figure C1.** Standardising *N*-gram lists

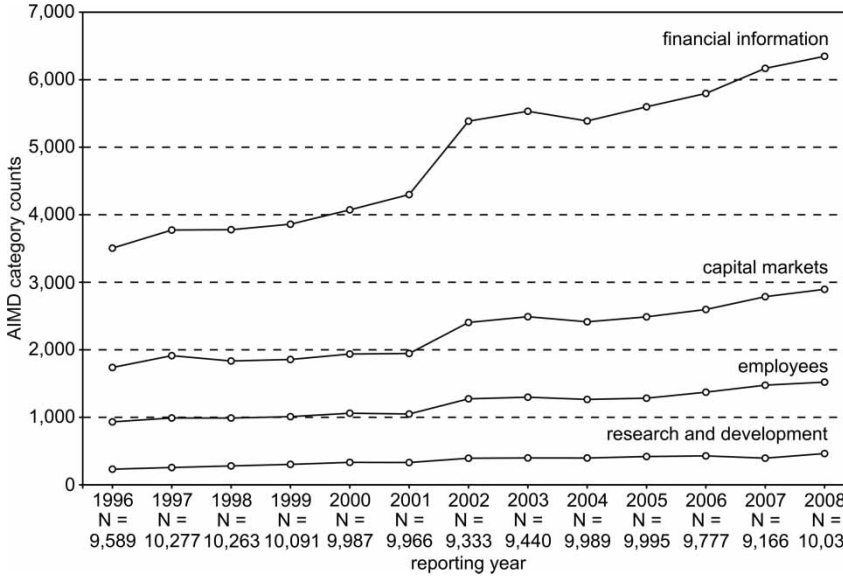



**Figure D1.** Average AIMD disclosure level for companies regulated by the SEC

The figure clearly reveals an increased level of disclosure by companies following the Sarbanes-Oxley Act of 2002. This could be interpreted as evidence for content validity of AIMD.

## Appendix E: Distribution of Disclosure within Individual Reports

Below, I illustrate how AIMD can be used to measure disclosure *within* an individual annual report, taking the DaimlerChrysler 2006 Annual Report as an example.

The word-based category counts of *N*-grams relating to a particular disclosure dimension are not equally distributed within individual reports. For example, some parts of a report provide more *financial information* while others focus on *social and environmental responsibility*. Figure E1 shows how *N*-grams relating to financial information are distributed between the seven chapters in the DaimlerChrysler 2006 Annual Report. Unsurprisingly, most financial information occurs in the financial statement and the management discussion.

Figure E2 shows that information relating to *social and environmental responsibility* likewise mainly occurs in the chapters dealing with these areas.

Similar curves (not reported) occur for the other disclosure dimensions, showing that information generally appears in relevant chapters of the report. This may also be seen as evidence for content validity of AIMD.

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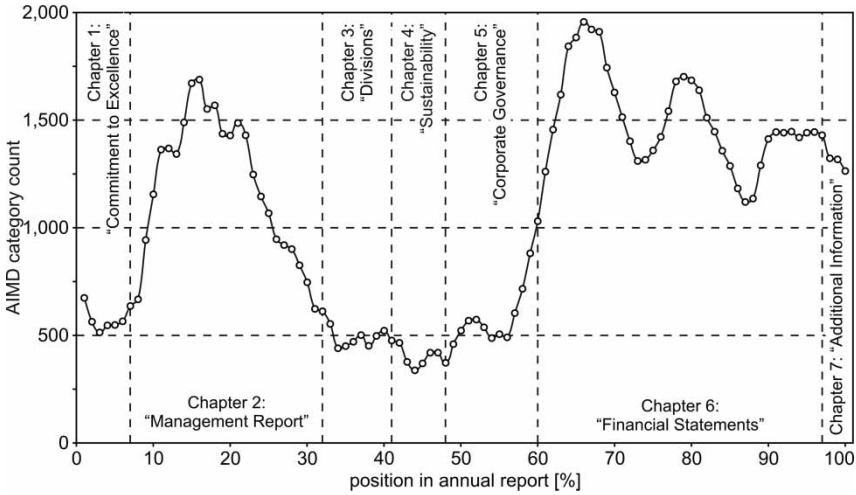



**Figure E1.** Distribution of AIMD category counts relating to *financial information* within the DaimlerChrysler 2006 Annual Report

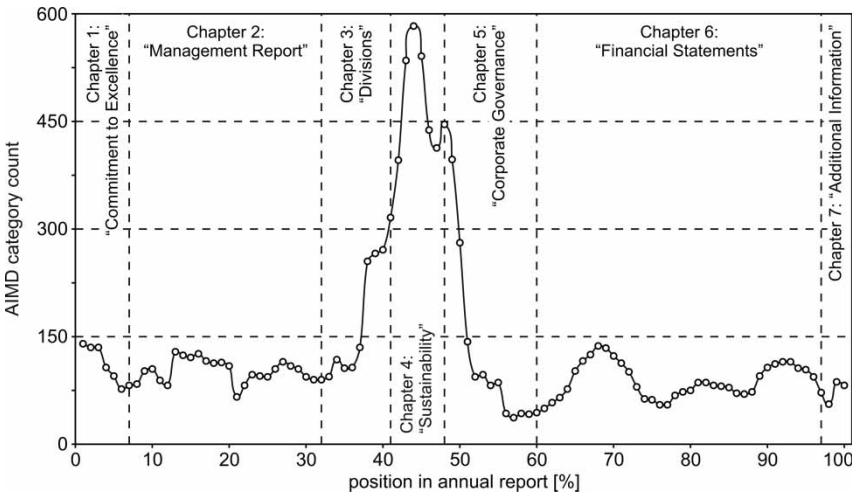



**Figure E2.** Distribution of AIMD category counts relating to *social and environmental responsibility* in the DaimlerChrysler 2006 Annual Report

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## Notes

[1]Studies claiming to measure the quality of disclosure use characteristic terms which are thought to indicate forward-looking, non-financial and non-quantitative information, for example.

[2]Alternatively convergence can also be measured using confirmative factor analysis or structural equation analysis. These techniques are identical for the two variable problems examined here.

[3]Although the rating covers around 200 companies, for most years results are publicly available for the top 50–110 companies only.

[4]Some firms used US GAAP during a transitional period.

[5]Financial institutions and insurance companies were excluded as they may use different wording and provide different content from manufacturing companies and service providers in their annual reports. Additional research is required into the heterogeneity or homogeneity of corporate documents with regard to particular sectors.

[6]For instance, the abbreviation 'EVA' appears on the financial information *N*-gram list while the Christian name 'Eva' is part of the residual category which does not relate to corporate disclosure.

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